

1 In cells, energy can be released from substrates by anaerobic and aerobic respiration.

(a) Most eukaryotic cells can respire anaerobically.

(i) Which of the following describes a step in the process of anaerobic respiration?

(1)

- ☐ **A** decarboxylation of lactate
- ☐ **B** phosphorylation of hexoses
- ☐ **C** oxidation of pyruvate
- ☐ **D** removal of phosphate groups from glucose

(ii) What happens to the lactate concentration during a period of anaerobic respiration?

(1)

- ☐ **A** decreases, causing a decrease in blood pH
- ☐ **B** decreases, causing an increase in blood pH
- ☐ **C** increases, causing a decrease in blood pH
- ☐ **D** increases, causing an increase in blood pH

(b) Most eukaryotic cells are also able to respire aerobically.

(i) How do respiratory substrates enter the Krebs cycle?

(1)

- ☐ **A** as molecules containing 2 carbon atoms produced by the link reaction
- ☐ **B** as molecules containing 3 carbon atoms produced by the link reaction
- ☐ **C** as molecules containing 2 carbon atoms produced by RUBISCO
- ☐ **D** as molecules containing 3 carbon atoms produced by RUBISCO

(ii) Draw a diagram of a mitochondrion.

Label only the part of the mitochondrion where the Krebs cycle occurs.

(2)

(iii) Describe the role of chemiosmosis in the synthesis of ATP.

(5)

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(Total for Question 1 = 10 marks)